

PEMBRY SANDS, United Kingdom

WMO: 036050

Lat: 51.7144N

Lon: 4.3675W

Elev: 7

StdP: 101.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.3	-1.9	-6.0	2.3	-0.8	-4.5	2.6	-0.1	21.9	8.7	18.9	8.7	2.4	70	0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	23.2	18.0	21.5	17.2	20.1	16.6	18.9	21.8	18.0	20.0	17.3	19.2	4.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.8	19.6	17.2	12.3	18.8	16.6	11.8	18.2	53.4	21.9	50.7	20.2	48.8	19.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.3	13.5	DB	-6.3	27.0	1.8	2.2	-7.5	28.6	-8.6	29.9	-9.6	31.2	-10.9	32.8	(4)
(5)			WB	-6.8	21.1	1.9	1.9	-8.1	22.5	-9.2	23.6	-10.2	24.7	-11.6	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	5.6	5.6	6.8	8.8	11.5	14.2	16.0	15.9	14.2	11.6	8.4	6.3	(6)	
	DBStd	4.67	3.08	2.77	2.60	2.57	2.51	2.20	2.02	2.01	2.26	2.90	3.19	3.56	(7)	
	HDD10.0	629	136	124	102	51	12	1	0	0	1	18	66	119	(8)	
	HDD18.3	2894	393	357	358	286	212	127	76	78	126	209	299	373	(9)	
	CDD10.0	789	1	0	2	14	59	126	187	184	126	68	17	4	(10)	
	CDD18.3	12	0	0	0	0	1	2	5	3	1	0	0	0	(11)	
	CDH23.3	57	0	0	0	0	2	11	27	16	1	0	0	0	(12)	
	CDH26.7	7	0	0	0	0	0	1	4	3	0	0	0	0	(13)	
(14) Wind	WSAvg	6.2	6.8	6.6	6.0	5.6	5.8	5.7	5.7	5.9	6.0	6.6	6.5	6.8	(14)	
(15) Precipitation	PrecAvg	1217	128	93	82	70	77	79	81	104	90	141	142	133	(15)	
	PrecMax	1462	223	186	162	139	156	189	219	202	175	237	263	229	(16)	
	PrecMin	967	14	31	37	21	18	29	15	21	30	42	59	45	(17)	
	PrecStd	142	52	45	32	36	32	41	48	46	41	54	52	52	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	12.2	16.4	19.6	22.7	24.7	26.3	25.7	22.8	18.7	15.0	13.0	(19)	
		MCWB	10.4	9.3	11.3	13.5	16.1	18.3	19.2	20.5	18.3	15.7	13.5	12.0	(20)	
	2%	DB	10.8	10.5	12.9	16.2	19.7	21.7	23.1	22.3	20.3	17.1	14.0	12.2	(21)	
		MCWB	10.0	9.1	9.9	11.8	14.9	17.0	18.1	18.0	16.7	15.0	13.1	11.3	(22)	
	5%	DB	10.2	9.8	11.4	14.2	17.4	19.8	21.3	20.5	19.0	16.1	13.4	11.6	(23)	
		MCWB	9.4	8.7	9.1	10.6	13.6	16.0	17.4	16.9	15.8	14.7	12.3	10.8	(24)	
	10%	DB	9.7	9.3	10.3	12.7	15.8	18.2	19.9	19.3	17.9	15.5	12.7	11.0	(25)	
		MCWB	8.9	8.3	8.7	10.1	12.9	15.1	16.7	16.3	15.6	14.1	11.6	10.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.4	12.1	14.2	17.3	19.6	20.0	21.0	18.8	16.6	14.2	12.4	(27)	
		MCDB	11.2	10.9	14.8	18.6	21.1	23.1	24.1	24.7	21.2	17.9	14.3	12.6	(28)	
	2%	WB	10.2	9.7	10.6	12.4	15.5	17.6	18.8	18.7	17.6	15.8	13.3	11.7	(29)	
		MCDB	10.5	10.1	12.2	15.0	18.6	20.7	22.0	21.0	19.2	16.5	13.7	12.1	(30)	
	5%	WB	9.7	9.1	9.8	11.4	14.3	16.6	18.0	17.8	16.9	15.1	12.7	11.1	(31)	
		MCDB	10.0	9.5	10.7	13.2	16.6	19.1	20.3	19.2	18.2	15.8	13.2	11.5	(32)	
	10%	WB	9.1	8.6	9.2	10.7	13.3	15.7	17.2	17.2	16.2	14.5	11.9	10.4	(33)	
		MCDB	9.6	9.1	10.0	12.1	15.2	17.6	19.2	18.6	17.5	15.3	12.5	10.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.4	6.3	7.2	7.3	6.9	6.6	6.6	7.0	6.1	5.7	5.1	(35)	
		MCDBR	3.9	5.1	7.8	9.6	10.5	10.7	10.0	8.8	8.8	5.5	4.4	3.8	(36)	
	5% WB	MCWBR	4.1	4.4	5.7	6.2	6.6	6.8	6.2	5.7	6.1	4.3	4.2	4.1	(37)	
		MCWBR	3.9	4.1	6.0	7.8	9.0	9.0	8.2	6.7	6.7	4.4	3.8	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.333	0.345	0.374	0.389	0.397	0.385	0.388	0.372	0.365	0.355	0.331	0.322	(40)	
		taud	2.331	2.373	2.313	2.298	2.319	2.380	2.377	2.438	2.447	2.433	2.386	2.308	(41)	
	Ebn at Noon		661	770	815	848	857	870	860	857	820	750	661	602	(42)	
		Edh at Noon	65	81	103	117	120	114	113	101	90	76	61	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.74	1.50	2.73	4.34	5.33	5.80	5.30	4.42	3.27	1.82	0.92	0.56	(44)	
	RadStd		0.07	0.14	0.25	0.33	0.33	0.50	0.45	0.30	0.24	0.15	0.08	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page